

本科毕业论文（设计）

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局部多孔质气体静压轴承关键技术的研究

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RESEARCH ON KEY TECHNOLOGIES OF PARTIAL POROUS EXTERNALLY PRESSURIZED GAS BEARING

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(Times New Roman 2号字加粗，题目太长时可用小2号字)



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哈尔滨工业大学← (楷体小2号字加粗)

2022年6月← (年、月用阿拉伯数字，
宋体、Times New Roman小2号字加粗)

□毕业论文 □毕业设计

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密级：公开

本科毕业论文（设计）



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局部多孔质气体静压轴承关键技术研究



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冒号左侧
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本 科 生：□□□
学 号：□□□□□□□□□□
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学 院：机电工程学院
答 辩 日 期：2022 年 6 月
学 校：哈尔滨工业大学

ZHANG San

Final Qualifying Work

XX

XX

Level of education: bachelor`s degree

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Wang Wu

Abstract

This is a sample abstract. Replace this paragraph with a concise description of the research problem, methods, main results, and conclusions. The abstract page uses the same font and line-spacing settings as the thesis body.

The template supports mathematical expressions, such as

$$\min_{x \in \mathbb{R}^d} F(x) := f(x) + g(x),$$

but the abstract should normally remain brief and readable.

Keywords: keyword one, keyword two, keyword three

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Chapter 1 Introduction

1.1 Background

This chapter demonstrates ordinary paragraphs, citations, equations, figures, tables, and section levels. Superscript citations are produced by `\cite`, for example^[1]. Inline citations are produced by `\inlincite`, for example [2].

The first paragraph after a heading is indented automatically. A displayed equation is numbered by chapter, as shown in equation (1-1).

$$\min_{x \in \mathbb{R}^d} F(x) := f(x) + g(x). \quad (1-1)$$

1.2 Figures

Figure 1.1 is a placeholder. Replace the framed box with `\includegraphics` when a real figure is available.



Figure 1.1 A placeholder figure

1.3 Tables

Table 1.1 shows a standard three-line table.

Table 1.1 Basic notation

Symbol	Meaning
x	Decision variable
f	Smooth function
g	Nonsmooth function

1.4 Heading Levels

1.4.1 Subsection Example

This is a subsection example.

1.4.1.1 Subsubsection Example

This is a subsubsection example.

Chapter 2 Methods

2.1 Algorithm Table

Algorithm 2.1 demonstrates the thesis-style algorithm table. The algorithm counter is independent from the table counter.

Algorithm 2.1 Sample algorithm

Input Initial point x_0 , stepsize γ , and maximum iteration number K .

Step 1 For $k = 0, 1, \dots, K - 1$, repeat the following computations:

1) Compute the gradient $\nabla f(x_k)$;

2) Update $x_{k+1} = x_k - \gamma \nabla f(x_k)$;

3) Check the stopping criterion.

Output x_K .

2.2 Theorem-like Environments

Definition 2.1 (Convex function). A function $f : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ is convex if, for any $x, y \in \text{dom } f$ and any $t \in [0, 1]$,

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y). \quad (2-1)$$

Lemma 2.1. *If f is convex, then every local minimizer of f is also a global minimizer.*

Proof. This follows from the definition of convexity. □

Theorem 2.1. *Under suitable assumptions, the sequence generated by Algorithm 2.1 is well defined.*

Proof. Replace this sample proof with the actual proof. □

Remark 2.1. The theorem, lemma, definition, assumption, proposition, corollary, example, and remark environments are numbered independently by chapter.

Chapter 3 Numerical Examples

3.1 Experimental Setting

This chapter demonstrates lists, subfigures, and numerical-result tables. Replace the sample text and placeholders with actual experimental settings and results.

- Dataset or test problem: describe the source and basic scale.
- Baseline methods: list the compared methods.
- Metrics: describe error, accuracy, time, or other evaluation metrics.

3.2 Subfigures

Figure 3.1 shows two placeholder subfigures.

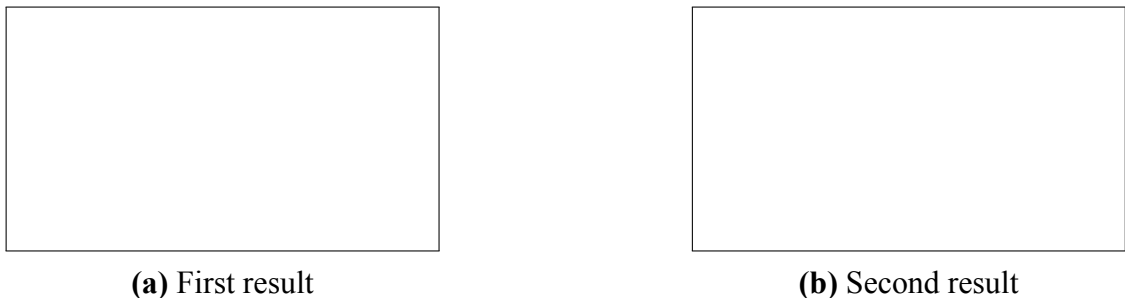


Figure 3.1 A sample subfigure layout

3.3 Result Table

Table 3.1 shows a sample result table.

Table 3.1 Sample numerical results

Method	Metric 1	Metric 2
Method A	0.123	1.23
Method B	0.098	1.05

Conclusion

This sample conclusion briefly summarizes the work completed in the thesis. Replace this paragraph with a concise discussion of the main contributions, verification results, limitations, and possible future work.

References

- [1] AUTHOR A, WRITER B. A sample journal article with protected ABC acronym[J]. Journal Name, 2026, 1(1): 1-10.
- [2] RESEARCHER C. A sample book title[M]. City: Publisher Name, 2026.

Appendix A: Basic Definitions and Theorems

A.1 Basic Definitions

This appendix demonstrates appendix numbering. Appendix sections are numbered as A.1, A.2, and so on. Appendix equations are numbered as equation (A-1).

$$\|x + y\| \leq \|x\| + \|y\|. \quad (\text{A-1})$$

A.2 Basic Theorems

Theorem A.1. *For any normed vector space, the triangle inequality holds.*

Proof. This is a sample appendix proof. □

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一、发表的学术论文

无

二、参与的科研项目及获奖情况

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致 谢

本论文的完成离不开许多老师与同学的关心与帮助。在此谨致以诚挚的感谢。